Abjad AI at NADI 2026 shared task: Overcoming Data Scarcity in Arabic Speech Processing through Augmentation

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Abstract

We describe our submission to three NADI 2026 subtasks: Tunisian Arabic–French–English code-switched ASR (Task 1.3), dialectal Arabic speech translation, and 20-way Arabic dialect identification (ADI). For code-switched ASR, we finetune a 2B-parameter FastConformer encoder-decoder with synthetic Tunisian speech, cross-lingual data, waveform augmentation, and checkpoint averaging, achieving 14.12% WER and ranking first overall. For speech translation, finetuning Whisper Large-v3 yields strong in-domain results but poor generalization (BLEU 5.77 on blind test), illustrating overfitting under limited data. For ADI, a classifier on frozen pretrained ASR representations outperforms full/partial finetuning, reaching 42% accuracy across 20 dialects. We highlight synthetic augmentation, checkpoint averaging, and encoder freezing as effective low-resource strategies.

1 Introduction

Arabic dialectal variation is compounded in North Africa by pervasive code-switching with French and English, and labeled speech resources for this setting remain scarce, making data efficiency central to system design. NADI 2026 (Sullivan et al., 2026) continues the NADI shared task series (Talafha et al., 2025), evaluating dialectal Arabic speech and language processing. We describe submissions to three speech subtasks: (1) Task 1.3, code-switched ASR for Tunisian Arabic–French–English; (2) dialectal Arabic speech translation; and (3) 20-way Arabic Dialect Identification (ADI). We adapt a 2B-parameter FastConformer (Rekesh et al., 2023) ASR model and Whisper Large-v3 using strategies suited to each task: full finetuning wins for the generation tasks (ASR, translation), while a frozen encoder with a lightweight classifier wins for ADI. Our ASR system ranked **first** on Task 1.3 (WER 14.12%); our ADI system reaches 42% accuracy; our translation system, despite strong training BLEU (44.6), generalizes poorly to blind test (BLEU 5.77), which we analyze as a negative result for low-resource speech translation.

2 Related Work

Arabic dialect identification Text-level dialect ID has been studied extensively via the NADI series (Abdul-Mageed et al., 2020; Talafha et al., 2025), while speech-based ID traces to the MGB/ADI17 challenge (Shon et al., 2020). ADI-20 (Elleuch et al., 2025a) extends ADI17 to 20 classes. Elleuch et al. (2025b) finetune Whisper-large-v3 for the coarser 8-way NADI 2025 task (79.83% accuracy); we instead freeze the encoder, which we find preferable for finer-grained 20-way classification.

Code-switched ASR Code-switching remains challenging for end-to-end ASR (Agro et al., 2026). Ben Abdallah et al. (2024) release TunSwitch, a Tunisian code-switched corpus, exploring self-supervised approaches. We instead fully finetune a large attention-based encoder–decoder model with a 48-layer FastConformer (Rekesh et al., 2023) encoder and an 8-layer Transformer decoder, using synthetic and cross-lingual data. Further details are provided in Section 3.1.

Automatic Speech Translation The IWSLT Dialectal and Low-Resource track (Anastasopoulos et al., 2022) established Tunisian-English speech translation as a low-resource benchmark; we similarly find parallel data scarcity, not model choice, is the binding constraint. Section 3.2 discusses our approach in details.

Frozen vs. Finetuned Representations SUPERB (Yang et al., 2021) showed frozen encoders competitive for tasks like language ID. We confirm this for ADI but find full finetuning wins for our generation tasks as reported in Section 4.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 47–52
October 28-29, 2026 ©2026 Association for Computational Linguistics

3 System Descriptions

3.1 (Task 1.3) Code-Switched ASR

We fully finetune Cohere Transcribe[1] model (Cohere, 2026a), a 2B-parameter encoder–decoder ASR model, on a Tunisian Arabic–French–English code-switched speech dataset provided by Task 1.3.

3.1.1 Model

An encoder-decoder model with a 48-layer Fast-Conformer (Rekesh et al., 2023) encoder and an 8-layer Transformer decoder with a 16,384-token vocabulary. It is a finetuned, Arabic-specialized derivative of Cohere Transcribe[2] model (Cohere, 2026b), adapted for dialectal variation and code-switching (Cohere, 2026a), which we further finetuned for Task 1.3.

3.1.2 Training

FP32 master weights with bfloat16 mixed-precision autocasting (loading directly in bfloat16 rounded small AdamW updates to zero). We use AdamW (Loshchilov and Hutter, 2019) with a cosine schedule and ∼3% linear warmup, on 3–5 A100 80GB GPUs via PyTorch Lightning (Falcon and The PyTorch Lightning team, 2019) DDP (Table 1). All 48 encoder and 8 decoder layers remain unfrozen throughout: full finetuning consistently outperforms partial/frozen configurations for this task, a pattern also holding for speech translation. Details are given in Section 3.2.

3.1.3 Data

Training combines the official NADI 2026 Task 1.3 data with: synthetic Tunisian speech via the pipeline in Figure 1 (transcripts rewritten by Gemma 4 (Gemma Team, 2026), synthesized by OmniVoice (Zhu et al., 2026) across 25 voices, with template artifacts removed and stronger augmentation than real recordings; and 9h French (Pratap et al., 2020) + 9h English (Panayotov et al., 2015) to improve non-Arabic segment recognition. Table 1 gives per-checkpoint data composition.

3.1.4 Augmentation

We explored two augmentation strategies in separate experiments. In the first, we applied *online* augmentations on-the-fly during training using the `audiomentations` library[3] (Jordal, 2022),

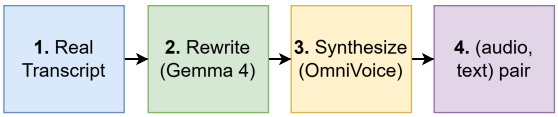


Figure 1: Synthetic data pipeline: a real transcript is rewritten by Gemma 4 for lexical variety, then synthesized by OmniVoice across 25 voices, to augment Task 1.3 training data.

with four methods: Gaussian noise, time stretching, pitch shifting, and gain perturbation, whose parameters varied per epoch. In the second, we applied *offline* augmentations, pre-generating augmented copies before training. Here we used the same four methods together with four additional ones: temporal shifting, band-pass filtering, tanh distortion, and ESC-50 (Piczak, 2015) background noise, creating five augmented copies per utterance.

3.1.5 Checkpoint Averaging

Our submission uniformly averages the weights of five checkpoints drawn from two experiments: online augmentation (CP1, CP2, CP5) and offline augmentation (CP3, CP4). These are not five independent runs: CP2 and CP5 are two checkpoints of the same online run, CP3 and CP4 are likewise two checkpoints of the same offline run (the later one taken after synthetic data was added to the training manifest), and CP1 was initialized from CP5 so the five checkpoints stem from three training runs in total (Table 1). Of these three runs, only the one producing CP1 is warm-started, from CP5's weights; the other two start from the original pretrained base model (Cohere, 2026a).

Weights are averaged in FP32 and cast back, following Izmailov et al. (2018). Normalization parameters and statistics (LayerNorm (Ba et al., 2016) and BatchNorm (Ioffe and Szegedy, 2015)), along with `num_batches_tracked`, are not averaged but taken from the best individual checkpoint among those combined.

As reported in Table 2, progressive averaging lowers WER from 16.64–16.80% for individual checkpoints to 15.97% for the 5-way average, and CER from 6.63–6.87% to 6.31%. The gain comes from combining the two experiments rather than from averaging alone: the two online checkpoints together (CP1, CP2) reach only 16.77% WER no better than the best single checkpoint (16.64%) whereas adding the first offline checkpoint (CP3) drops WER to 16.05%.

[1] https://huggingface.co/CohereLabs/cohere-transcribe-arabic-07-2026

[2] https://huggingface.co/CohereLabs/cohere-transcribe-03-2026

[3] https://github.com/iver56/audiomentations

#	Training Configuration	Run	Init. Weights	LR	Aug.	Val. WER (%)
1	Real + synthetic + 9h French + 9h English	R2	From CP 5	1×10^{-5}	Online	16.64
2	Real only (early ckpt)	R1	Base	5×10^{-6}	Online	16.80
3	Real + offline aug. (early ckpt)	R3	Base	5×10^{-6}	Offline	16.80
4	+ synthetic (later ckpt)	R3	Base	5×10^{-6}	Offline	16.67
5	Real only (later ckpt)	R1	Base	5×10^{-6}	Online	16.67

Table 1: The five checkpoints combined in our final submission, with their training configuration and validation WER on the NADI 2026 Task 1.3 development set, prior to averaging.

	WER (%)		CER (%)	
Team	**Raw**	**Norm.**	**Raw**	**Norm.**
Abjad AI (ours)	**14.41**	**14.12**	5.55	5.43
Itgan	14.49	14.22	**5.38**	**5.27**
Thaka	14.53	14.26	6.34	6.26

Table 3: NADI 2026 Task 1.3 (Code-Switched ASR) official blind test leaderboard, top three teams. *Norm.* denotes Hamza normalization.

	Configuration	WER	CER
Online	CP1: Real + synth. + FR/EN	16.64	6.63
	CP2: Real only	16.80	6.87
	CP5: Real only	16.67	6.68
Offl.	CP3: base	16.80	6.83
	CP4: + synth.	16.67	6.69
Avg(1,2)		16.77	6.82
Avg(1,2,3)		16.05	6.39
Avg(1,2,3,4)		16.05	6.42
Avg(1,2,3,4,5) [final]		**15.97**	**6.31**

Table 2: WER/CER (%) on the NADI 2026 Task 1.3 development set for the five individual checkpoints and their progressive averages. The final row is the 5-way average submitted on the blind test set.

3.1.6 Results

The submitted system ranked **first** on the NADI 2026 Task 1.3 blind test set. The achieved performance scores are reported in Table 3[4]. Synthetic speech, multilingual data, diverse augmentation, and checkpoint averaging together substantially improve code-switched ASR under limited-data conditions.

3.2 (Task 4) Speech Translation

We finetune Whisper Large-v3 (Radford et al., 2023) (1.55B parameters) on the shared-task data for dialectal Arabic speech translation.

3.2.1 Training Strategy

Direct finetuning on the shared-task dataset consistently outperformed (3.98 vs 5.77 Eval BLEU score) a two-stage approach that first finetuned on an internal, out-of-domain Arabic Multi-dialects dataset of 800 hours before finetuning on the shared-task data. The domain mismatch likely outweighed the benefit of additional examples.

[4] https://www.codabench.org/competitions/17167/

Split	# Samples	Hours	BLEU
Training set	8644	10.18	–
Dev (Val) set	680	0.79	21.75
Blind test set	6780	7.78	5.77

Table 4: BLEU for our best speech translation configuration (Whisper Large-v3 finetuned on shared-task data augmented for the three weakest dialects) on the NADI 2026 training, eval and blind test sets.

3.2.2 Data Augmentation

To reduce overfitting under limited data, we apply the same offline waveform augmentation (`audiomentations` (Jordal, 2022)) as in the previous task: Gaussian noise, time stretching, pitch shifting and temporal shifting. As before, augmented copies are pre-generated rather than applied on-the-fly. Rather than upsampling the training set, we target augmentation at the three weakest-performing dialects, generating acoustically varied copies for these subsets only. This targeted dev set contains **18/680 = 2.65%** overlap.

3.2.3 Training

We fully finetune Whisper Large-v3 with all layers unfrozen, consistent with Section 3.1, where fuller unfreezing improved performance. Our strongest configuration finetuned directly on the shared-task data augmented with the targeted three-dialect set, with no intermediate stage.

3.2.4 Results

The large train/test gap indicates substantial overfitting: targeted augmentation improves fit to limited data but lacks the diversity needed for generalization. Future work should prioritize collecting

or generating genuinely diverse dialectal speech-translation data over duplicated, acoustically modified examples.

3.3 (Task 2) Arabic Dialect Identification (ADI)

We formulate ADI as 20-class utterance-level classification (MSA + 19 dialects). Our final system uses frozen acoustic representations from the same Cohere ASR encoder (Cohere, 2026a) used in Section 3.1, with a lightweight trainable classifier.

3.3.1 Model

Frame-level hidden states are mean-pooled per utterance, projected to 256 dimensions (linear + layer norm), then passed through a classification head (linear, GELU, dropout, linear) producing logits over 20 classes. The encoder remains fully frozen; only the projection and classifier are trainable. This outperformed partial or full finetuning on validation data, the opposite of the pattern for ASR and speech translation (see Section 4 for details).

3.3.2 Data and Preprocessing

We train on ADI-20 dataset; most of the official validation set is merged into training, with a small stratified subset retained for model selection. Audio is converted to mono channel, resampled to 16kHz, and RMS-normalized to discourage reliance on recording volume as a shortcut.

3.3.3 Data Augmentation and Training

Waveform augmentation (random gain, additive noise, time shifting, speed perturbation, simulated bandwidth reduction) improves robustness. We use class-balanced cross-entropy with AdamW (Loshchilov and Hutter, 2019), cosine annealing, and bfloat16 mixed precision on multiple GPUs.

3.3.4 Alternative Systems

We also explored GRU-based classification heads on Qwen3-ASR (Shi et al., 2026) and Whisper Large-v3 (Radford et al., 2023) representations. For Qwen3-ASR, a frozen encoder with augmented-only training performed best; unfreezing consistently hurt blind-test performance. For Whisper, we additionally tried intermediate training on VoxLingua107 (Valk and Alumäe, 2021), checkpoint averaging, and a zero-shot multimodal-instruction-model baseline. None outperformed the frozen Cohere-based system.

3.3.5 Results

Our system reaches **42.0%** accuracy on the blind test set, well above the 5.0% random-chance baseline for 20-way classification. Under ADI-20's limited-data conditions, combining strong pretrained Arabic speech representations with augmentation and a small frozen-encoder classifier consistently generalizes better than full encoder finetuning.

4 Discussion

Whether full finetuning or a frozen encoder generalizes better depends on task type, not backbone. For generation tasks – ASR and speech translation – unfreezing the full encoder/decoder consistently wins; for ADI, with less labeled data per class, a frozen encoder with a lightweight classifier wins across every backbone. Generation tasks likely benefit from adapting the acoustic-to-text mapping, while limited-data classification is more prone to overfitting when fully finetuned.

Augmentation and synthetic/cross-lingual data help, but only atop a reasonably sized authentic pool: they boosted our top-ranked ASR system, yet failed to prevent severe overfitting for speech translation's ∼2.5-hour pool – augmentation multiplies existing diversity rather than replacing it. Checkpoint averaging gave a further consistent WER/CER gain as indicated in Table 2.

5 Conclusion

We presented submissions to three NADI 2026 subtasks. Our code-switched ASR system, combining synthetic/cross-lingual data, augmentation, and checkpoint averaging, ranked first on Task 1.3 (WER 14.12%). Our ADI system shows a frozen encoder with a lightweight classifier generalizes better under limited data (42% accuracy). Our speech translation system, by contrast, illustrates overfitting under an extremely small training pool (BLEU score of 5.77).

Limitations

Our ASR system incorporates synthetic Tunisian TTS speech; despite artifact removal and stronger augmentation, some mismatch with authentic speech likely remains and its isolated contribution is unquantified. Four of five averaged checkpoints come from three training runs: two checkpoints are from one run, two from another, and the fifth was initialized from one of those checkpoints, so

the averaging benefit may partly reflect correlated checkpoints. Our speech translation system generalizes poorly (BLEU 5.77 vs. 44.58 train) and should not be taken as representative of achievable performance with more data. Our ADI backbone comparison (Cohere, Qwen3-ASR, Whisper) was not run under identical compute/tuning budgets, so rankings are suggestive; two of our three systems (ASR and ADI) share the same underlying encoder, so this comparison offers weaker evidence for a general task-type effect than fully independent backbones would. The Task 2 blind test set does not provide per-dialect labels, so we are unable to report a dialect-wise breakdown of our ADI system's errors.

References

Muhammad Abdul-Mageed, Chiyu Zhang, Houda Bouamor, and Nizar Habash. 2020. NADI 2020: The first nuanced Arabic dialect identification shared task. In *Proceedings of the Fifth Arabic Natural Language Processing Workshop*, pages 97–110, Barcelona, Spain (Online). Association for Computational Linguistics.

Maha Tufail Agro, Atharva A. Kulkarni, Karima Kadaoui, Zeerak Talat, and Hanan Aldarmaki. 2026. Code-switching in end-to-end automatic speech recognition: A systematic literature review. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 9790–9812, Palma de Mallorca, Spain. ELRA Language Resource Association.

Antonios Anastasopoulos, Loïc Barrault, Luisa Bentivogli, Marcely Zanon Boito, Ondřej Bojar, Roldano Cattoni, Anna Currey, Georgiana Dinu, Kevin Duh, Maha Elbayad, Clara Emmanuel, Yannick Estève, Marcello Federico, Christian Federmann, Souhir Gahbiche, Hongyu Gong, Roman Grundkiewicz, Barry Haddow, Benjamin Hsu, and 24 others. 2022. Findings of the IWSLT 2022 evaluation campaign. In *Proceedings of the 19th International Conference on Spoken Language Translation (IWSLT 2022)*, pages 98–157, Dublin, Ireland (in-person and online). Association for Computational Linguistics.

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. 2016. Layer normalization. *arXiv preprint arXiv:1607.06450*.

Ahmed Amine Ben Abdallah, Ata Kabboudi, Amir Kanoun, and Salah Zaiem. 2024. Leveraging data collection and unsupervised learning for code-switched Tunisian Arabic automatic speech recognition. In *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 12607–12611. IEEE.

Cohere. 2026a. Cohere transcribe arabic: Frontier speech recognition for Arabic. Cohere Blog. Accessed August 2026.

Cohere. 2026b. Cohere transcribe: State-of-the-art speech recognition. Cohere Blog. Accessed August 2026.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025a. ADI-20: Arabic dialect identification dataset and models. In *Proc. Interspeech 2025*, pages 2775–2779.

Haroun Elleuch, Youssef Saidi, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025b. ELYADATA & LIA at NADI 2025: ASR and ADI subtasks. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 762–766, Suzhou, China. Association for Computational Linguistics.

William Falcon and The PyTorch Lightning team. 2019. PyTorch Lightning. GitHub.

Gemma Team. 2026. Gemma 4 technical report. *Preprint*, arXiv:2607.02770.

Sergey Ioffe and Christian Szegedy. 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 448–456, Lille, France. PMLR.

Pavel Izmailov, Dmitrii Podoprikhin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. 2018. Averaging weights leads to wider optima and better generalization. In *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence (UAI)*.

Iver Jordal. 2022. audiomentations: A Python library for audio data augmentation. GitHub.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: An ASR corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5206–5210. IEEE.

Karol J. Piczak. 2015. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM International Conference on Multimedia*, pages 1015–1018. ACM.

Vineel Pratap, Qiantong Xu, Anuroop Sriram, Gabriel Synnaeve, and Ronan Collobert. 2020. MLS: A large-scale multilingual dataset for speech research. In *Proc. Interspeech 2020*, pages 2757–2761.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning*, pages 28492–28518. PMLR.

Dima Rekesh, Nithin Rao Koluguri, Samuel Kriman, Somshubra Majumdar, Vahid Noroozi, He Huang, Oleksii Hrinchuk, Krishna Puvvada, Ankur Kumar, Jagadeesh Balam, and Boris Ginsburg. 2023. Fast conformer with linearly scalable attention for efficient speech recognition. In *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–8. IEEE.

Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, Jin Xu, Jingren Zhou, and Junyang Lin. 2026. Qwen3-ASR technical report. *arXiv preprint arXiv:2601.21337*.

Suwon Shon, Ahmed Ali, Younes Samih, Hamdy Mubarak, and James Glass. 2020. ADI17: A fine-grained Arabic dialect identification dataset. In *ICASSP 2020 – 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8244–8248. IEEE.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, Abdel-Rahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. NADI-2026: The Second Multidialectal Arabic Speech Processing Shared Task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. NADI 2025: The first multidialectal Arabic speech processing shared task. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China. Association for Computational Linguistics.

Jörgen Valk and Tanel Alumäe. 2021. VoxLingua107: A dataset for spoken language recognition. In *2021 IEEE Spoken Language Technology Workshop (SLT)*, pages 652–658. IEEE.

Shu-wen Yang, Po-Han Chi, Yung-Sung Chuang, Cheng-I Jeff Lai, Kushal Lakhotia, Yist Y. Lin, Andy T. Liu, Jiatong Shi, Xuankai Chang, Guan-Ting Lin, Tzu-Hsien Huang, Wei-Cheng Tseng, Ko-tik Lee, Da-Rong Liu, Zili Huang, Shuyan Dong, Shang-Wen Li, Shinji Watanabe, Abdelrahman Mohamed, and Hung-yi Lee. 2021. SUPERB: Speech Processing Universal PERformance Benchmark. In *Interspeech 2021*, pages 1194–1198.

Han Zhu, Lingxuan Ye, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang, Zhifeng Han, Weiji Zhuang, Long Lin, and Daniel Povey. 2026. OmniVoice: Towards omnilingual zero-shot text-to-speech with diffusion language models. *arXiv preprint arXiv:2604.00688*.